

# TrustAI: An Expert-Based Methodology for Evaluating and Certifying the Trustworthiness of Artificial Intelligence Systems

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**Abstract**—Despite the acknowledged importance of ensuring the trustworthiness of artificial intelligence (AI) systems, there is currently a lack of well-established methods and techniques for evaluating and certifying their trust. To address this issue, we propose TrustAI, a novel expert-based methodology for evaluating and certifying the trustworthiness of AI systems. Our approach is based on a set of high-level requirements, i.e., the pillars of trust, which are decomposed into a set of low-level criteria expressed in the form of questions that are assessed by AI experts. It aggregates the expert judgments through a hierarchical model to provide a quantitative expression of the level of trust (LoT) of a given AI system and generates a certificate in case that the evaluation results are satisfactory. Fuzzy logic is utilized for modeling the uncertainty of expert judgments and for defining the parameters of the model. The proposed methodology is highly configurable, enabling the generation of custom evaluation models tailored to specific application domains and regulatory frameworks. It is illustrated through a use case on two actual AI systems, evaluated using a model derived via the methodology based on a recently issued regulatory framework.

**Impact Statement**—AI models evolve at a rapid speed, demonstrating remarkable capacity in performing everyday activities, often surpassing humans. This, along with their increasing adoption in practice, has raised concerns in society regarding the dangers of AI, which have been formally expressed by scientists, governments, and even by Geoffrey Hinton, the “Godfather of AI.” Validating AI trustworthiness is critical for ensuring that AI systems are safe, reliable, secure, and respect ethical values, thereby helping in building societal trust. Despite its importance, almost no concrete methods for AI trustworthiness evaluation exist, apart from some governmental tools that are regulation-specific and scientific initiatives, which are cryptic and non-operationalized in the form of tools. In this article, we propose TrustAI, an expert-based, regulation-agnostic methodology for evaluating and certifying AI trustworthiness. The methodology is highly configurable, being able to be adapted to specific regulations, standards, and domains, whereas it is available as a practical tool to be used and tested by practitioners and researchers. Our work is among the first (if not the first) to

provide an open, transparent methodology for evaluating AI trust, offering a concrete basis for researchers to build future models and frameworks.

**Index Terms**—Artificial intelligence (AI), certification, evaluation, expert systems, fuzzy logic, trust.

## I. INTRODUCTION

**D**URING the past years, artificial intelligence (AI) systems have started being a fundamental part of our daily lives, monitoring and handling critical everyday activities in various domains, from transportation to healthcare. Therefore, since their decisions may greatly affect individuals and organizations, their trustworthiness is a matter of major concern for society in general. The concerns about the trustworthiness of AI systems have become more prominent today, with the recent breakthroughs in large language models (LLMs), and particularly with the release of generative pre-trained transformer (GPT) [1]. These models have demonstrated remarkable humanlike capabilities in processing natural language, raising concerns about whether they are safe to use or may introduce inequalities or violate ethical values.

The first formal expression of these apprehensions was made through an open letter that was signed by renowned scientists and technology leaders, asking for a pause in the development of AI models more advanced than GPT-4 for at least six months so that their potential risks to society can be evaluated early.<sup>1</sup> Since then, several governments have expressed their concerns about the risks that are imposed by AI and the need to ensure their trust. Notably in 2023, former U.S. President Biden released an executive order requiring AI companies to assess national security risks before releasing their technology to the public, whereas the U.K. government released a document called the Bletchley Declaration,<sup>2</sup> signed by 28 governments, warning of the dangers posed by advanced AI systems and expressing the need for international collaboration on safety standards and regulatory frameworks for AI. Finally, Geoffrey Hinton, often referred to as the “Godfather of AI,” expressed his concerns about the potential risks of AI during his Nobel Prize acceptance speech, further contributing to societal apprehension about the widespread adoption of AI.<sup>3</sup>

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<sup>1</sup><https://futureoflife.org/open-letter/pause-giant-ai-experiments/>

<sup>2</sup><https://www.gov.uk/government/publications/ai-safety-summit-2023>

<sup>3</sup><https://www.nobelprize.org/prizes/physics/2024/hinton/speech/>

To alleviate such concerns, there is a need for evaluating and certifying the trustworthiness of AI systems, i.e., to assess whether these systems satisfy important requirements, such as accuracy, reliability, security, and transparency, among others. This would help society to build trust in AI systems, whereas it could also help their developers to identify and fix potential issues prior to their release on the market.

However, despite the acknowledged need to ensure the trustworthiness of AI systems, there is currently a lack of well-established evaluation and certification methods. This lack can be attributed to the fact that, until recently, there were no regulatory frameworks for AI and trust, resulting in a lack of a common standard that could set the basic logic of the assessment. Only recently have relevant attempts in regulating AI been made, with the most notable being the regulatory framework proposed by the European Union (EU) and voted in December 2023, known as the “AI Act” [2], being the first formal legislation and regulation for AI trustworthiness.

To this end, in the present article, we propose TrustAI, a novel methodology for evaluating and certifying the trustworthiness of AI systems. The proposed approach is based on a set of high-level trustworthiness requirements, i.e., the “pillars of trust” (PoT), which are decomposed into a set of low-level criteria, expressed in the form of questions, evaluated by AI experts. The expert judgments are aggregated through a hierarchical evaluation model (HEM), providing a quantitative expression of the level of trust (LoT) of the analyzed system and issuing a certificate if evaluation results are sufficient. Since the proposed methodology is based on expert judgments, fuzzy logic is used to model uncertainty. The proposed methodology is highly configurable, enabling the construction of custom models tailored to specific application domains and regulatory frameworks, mainly through a fuzzy multi-criteria decision-making (MCDM) approach. The methodology is demonstrated through a use case on two real-world AI systems, analyzed using a sample model derived from the artificial intelligence act (AIA) [2].

The only similar attempt is capAI [3], which enables the conduct of a conformity assessment against AIA. However, it is not configurable, focusing only on the AIA framework and particularly on the single criterion of Ethics. A tool is not available to evaluate it in practice, and no information about its internal structure is provided, which prevents its replication. In addition to this, recently, the European Commission’s high-level expert group on AI (AI HLEG) has made available the assessment list for trustworthy AI (ALTAI) [4], which is a self-assessment checklist that organizations can use to evaluate AI systems under development based on the main criteria defined by the AI Act. Hence, ALTAI is simply a checklist-based self-assessment allowing users to detect potential gaps in AI applications, being specific to a single regulatory framework, i.e., the AI Act. To our knowledge, TrustAI is the first expert-based regulatory-agnostic methodology to assess and certify the trustworthiness of AI systems, suitable for self-evaluation and third-party audits.

## II. MOTIVATION AND BACKGROUND

### A. Common Trustworthiness Criteria

An AI system should be evaluated and certified against a set of criteria, the selection of which depends mainly on: 1) the selected regulatory framework; and 2) the application scenario or domain (i.e., how critical the activities handled by the AI system are). The main and most frequently cited trustworthiness criteria (i.e., requirements), commonly considered during verification and validation of AI systems [2], [6], [7], [8], include the following.

- 1) *Accuracy*: It refers to the ability of an AI system to correctly classify information into categories, or make correct predictions, recommendations, or decisions [6].
- 2) *Reliability*: It refers to the ability of an AI system to work properly with a range of inputs and in a range of situations, producing results that can be reproduced, ensuring that it exhibits consistent behavior [6].
- 3) *Security*: It refers to the ability of an AI system to remain resilient against attempts by unauthorized third parties to alter its use, behavior, outputs, or performance by exploiting system vulnerabilities. This includes protecting against AI-specific threats such as data poisoning, model poisoning, and adversarial attacks [6].
- 4) *Privacy*: It refers to the ability of an AI system to guarantee data protection and privacy throughout its entire lifecycle, including both information initially provided by the user and information generated about them [6].
- 5) *Transparency*: It refers to the ability of the AI system to be developed and used in a way that allows appropriate traceability and explainability. It comprises three distinct elements: a) traceability; b) explainability; and c) open communication about the limitations of the AI system [6].
- 6) *Safety*: It refers to the ability of an AI system to perform its intended function without harming living beings or the environment [6].
- 7) *Fairness*: It refers to the ability of an AI system to ensure that individuals and minority groups remain free from bias, stigmatization, and discrimination [6].
- 8) *Human Agency and Oversight*: It refers to the ability of an AI system to support users in making informed autonomous decisions and to allow for human intervention through mechanisms such as human-in-the-loop, human-on-the-loop, and human-in-command to ensure the system does not undermine human autonomy [6].

It should be noted that these trustworthiness criteria are listed because they are among the most highly cited and commonly referred to criteria in the related literature [2], [6], [7], [8], while they are in line with the criteria and terminology of the AI Act, providing a stronger theoretical formulation. To further ground the trustworthiness dimensions considered in this work in an additional established and widely adopted standard (beyond the AI Act), we align the identified trustworthiness criteria (i.e., PoT) with the trustworthiness characteristics defined by the NIST Artificial Intelligence Risk Management Framework (AI

TABLE I  
NIST AI RMF [5] TRUSTWORTHINESS CHARACTERISTICS AND MAPPING TO TRUSTAI PILLARS

| NIST Characteristic           | Definition (Summary From NIST AI RMF 1.0)                                                                                            | TrustAI Pillar(s)             |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Valid and Reliable            | The system fulfills intended use requirements and operates correctly over time (reliability), supported by accuracy and robustness.  | Accuracy; Reliability         |
| Safe                          | The system must not endanger life, health, property, or environment; ensured through testing, monitoring, and human intervention.    | Safety                        |
| Secure and Resilient          | The system must resist adversarial manipulation and exfiltration, withstand, recover from adverse events, and degrade safely.        | Security                      |
| Accountable and Transparent   | The system must provide transparent information about itself to enable accountability, governance, and redress across the lifecycle. | Transparency                  |
| Explainable and Interpretable | The system explains underlying mechanisms (how) and interprets outputs (why) to support documentation, auditing, and governance      | Transparency / Explainability |
| Privacy-Enhanced              | The system safeguards autonomy, identity, and dignity, adhering to privacy norms, ensuring anonymity, confidentiality, and control.  | Privacy                       |
| Fair (Harmful Bias Managed)   | The system must manage systemic, statistical, and cognitive biases to ensure contextual equality and equity.                         | Fairness                      |

RMF) [5]. Table I summarizes the NIST taxonomy and maps each NIST characteristic to the corresponding PoT that we identified.

From Table I, it is evident that different AI trustworthiness standards, regulations, and evaluation frameworks, despite their high overlap, exhibit subtle but distinctive differences in the definition of various trustworthiness dimensions. For instance, in NIST's AI RMF, transparency and explainability are treated as separate criteria, whereas in the AI Act, they are merged together. A reasoning similar to NIST's AI RMF is also adopted by other recent research endeavors [9], [10], [11]. Similarly, the AI Act identifies human agency and oversight as an individual criterion, while in NIST AI RMF, it is considered a cross-cut element. This renders the construction of a unified evaluation model impossible and, in turn, stresses the need for highly adaptable and configurable assessment methodologies. TrustAI alleviates this issue by enabling the construction of custom assessment models tailored to specific standards and regulatory frameworks. For instance, TrustAI could be employed to assess the trust of a given AI system against the dimensions and guidelines of the NIST framework.

### B. Trustworthiness Criteria Interdependencies

According to [5], trustworthiness characteristics are interdependent, i.e., they influence each other and thus cannot be addressed in isolation. For instance, the "valid and reliable" characteristic is foundational for all other characteristics, since inaccurate and unreliable behavior will only amplify other downstream risks (e.g., safety, security, and fairness). In addition to that, the lack of "transparency" renders the models opaque, making it difficult to assess other requirements, such as their security, privacy, and fairness. Hence, often these interdependencies surface as trade-offs that vendors and reviewers need to take into consideration when building or evaluating these systems. These interdependencies and trade-offs are considered by the TrustAI methodology, both via the multicriteria aggregation model (Section III) and, chiefly via the proposed weights elicitation scheme (Section IV), enabling the model's weights to reflect the expert-perceived interdependencies of the criteria.

### C. Trustworthiness Criteria Validation

Validating the trustworthiness criteria requires manual effort and expertise [12]. *Fairness* involves *fairness testing* for detecting biases, and strong expertise for results interpretation. *Explainability* is assessed based on the model's inherent

transparency or the adoption of dedicated techniques such as SHapley Additive exPlanations (SHAP) [13]. *Security* and *privacy* require manual code inspection and GDPR compliance, respectively. *Reliability* and *accuracy* require repeated model execution on unknown data to ensure consistent decision-making. Finally, *safety* requires formal risk analysis.

From the above analysis, it is clear that the assessment of the individual criteria, and, in turn, the broader evaluation and certification of the trustworthiness of an AI system is a complex process that, in most cases, cannot be done automatically. It is based chiefly on audits performed by experts in AI. Therefore, there is a strong need for a methodology that could help experts measure the extent to which critical trustworthiness criteria are met by a given AI system. Since the evaluation is based mainly on expert judgments, an expert-based approach seems to be the most suitable solution.

## III. PROPOSED APPROACH

The purpose of the present work is to provide a methodology for evaluating the LoT of a given AI system in a quantitative manner, based on expert judgments. The high-level overview of the proposed methodology is illustrated in Fig. 1. As can be seen in Fig. 1, an AI expert provides their evaluations on a set of low-level criteria (expressed in the form of questions) through a dedicated questionnaire. Their answers are passed through a fuzzifier, which is responsible for turning their linguistic responses into fuzzy sets. The fuzzy responses are aggregated by a HEM and the overall LoT of the analyzed AI system is computed. Since the LoT is a fuzzy number, a defuzzification step is employed to compute a crisp value and a linguistic rank of the final assessment to enhance the understandability of the results. Finally, if the evaluation results are found to be sufficient, a certificate of trust is generated. Otherwise, a report with the identified issues is presented to the expert. The methodology is described in detail in the rest of this section.

### A. Evaluation Criteria and Linguistic Variables

As already stated, the evaluation of the LoT of a given AI application using the proposed methodology is based on expert judgments. Suppose that we want to evaluate the trustworthiness of an AI application based on  $N$  PoT, such as accuracy, reliability, and security, which were described in detail in Section II and are typically defined based on formal regulations

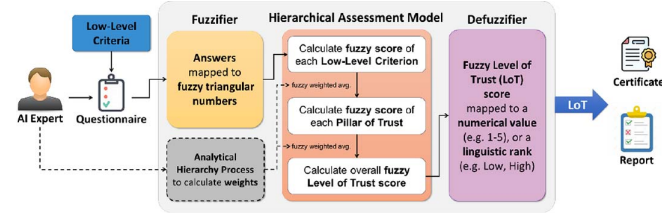


Fig. 1. High-level overview of the proposed AI trustworthiness evaluation and certification methodology.

such as the EU's AI Act [2]. Each one of the  $N$  pillars is decomposed into  $M$  low-level criteria that the experts have to assess, which are expressed in the form of individual questions. An indicative example of the high-level criteria and the main questions the experts have to answer for each pillar is illustrated in Table IV. Obviously, a different number of questions should be answered for each pillar.

The experts provide their responses in a linguistic form. The reasoning behind this decision is that linguistic values are more intuitive and easier to perceive by the human brain compared to numerical values. In particular, the expert needs to provide two answers: 1) their main answer that denotes the extent to which the low-level criterion is satisfied; and 2) how certain (i.e., confident) they are of their answer.

Let  $R_m$  be the variable that represents the main answer of the expert to a given low-level criterion (i.e., question).  $R_m$  takes its values from the set  $V = \{VL, L, M, H, VH\}$ , where  $VL$ ,  $L$ ,  $M$ ,  $H$ , and  $VH$  stand for *very low*, *low*, *medium*, *high*, and *very high*, respectively, and therefore  $R_m \in V$ . Similarly, let  $R_c$  be the linguistic variable that corresponds to the uncertainty of the expert's judgment.  $R_c$  receives a value from the set  $S = \{C, D, U\}$ , and therefore  $R_c \in S$ , where  $C$  denotes certain judgment,  $U$  uncertain judgment, and  $D$  corresponds to a default uncertainty. The latter value is used when the expert cannot define their uncertainty.

Hence, one of the benefits of the proposed methodology is that it enables the experts to express (if they consider it necessary) how certain they are for each one of their answers. This is important as the methodology allows to model the uncertainty that inevitably underlies expert judgments and takes it into account during the overall evaluation of the trustworthiness of the analyzed system.

### B. Fuzzifier

The purpose of the fuzzifier is to take the linguistic values provided by the expert for each individual question answered by them (i.e., answer along with its uncertainty) and assign a fuzzy set, particularly a fuzzy triangular number, to each one of these questions. A formal description of the fuzzification method adopted by the proposed approach is provided as follows.

Suppose that  $a$  is the fuzzy triangular set that corresponds to the response of an expert to a specific question (i.e., low-level criterion). To state it more formally,  $a$  is a triangular fuzzy number of the form  $a = (a_l, a_m, a_u)$  where  $a_l, a_m, a_u \in \mathbb{R}$ . For formally defining the membership function of the fuzzy

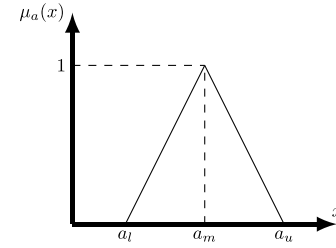


Fig. 2. Membership function of a fuzzy triangular number.

triangular number  $a$ , we use a special case of the definition of a fuzzy number provided by Dubois and Prade [14], as stated in [15]. This definition is provided as follows.

**Definition 1:** We define a fuzzy number  $a$  on  $R = (-\infty, +\infty)$  to be a triangular fuzzy number if its membership function  $\mu_a : R \rightarrow [0, 1]$  is equal to

$$\mu_a(x) = \begin{cases} \frac{x-a_l}{a_m-a_l}, & x \in [a_l, a_m] \\ \frac{a_u-x}{a_u-a_m}, & x \in [a_m, a_u] \\ 0, & x < a_l \text{ and } x > a_u \end{cases} \quad (1)$$

with  $a_l \leq a_m \leq a_u$ , where  $l$ ,  $m$ , and  $u$  stand for the lower, modal, and upper values of the triangular number, respectively. The support  $s$  is the set of elements  $\{x \in R | a_l < x < a_u\}$ . The triangular number  $a$  is depicted in Fig. 2, and we refer to it as  $a = (a_l, a_m, a_u)$ .

As can be seen by Fig. 2 and (1), to define a fuzzy triangular number, two parameters need to be set, which are: 1) its center (i.e., the value of  $a_m$ ); and 2) its range (i.e., the difference  $a_u - a_l$ ). These parameters are defined based on the two linguistic values that are provided by the experts (see Section III-A). More specifically, the center  $a_m$  of the triangular number is defined based on the main response  $R_m$  of the expert as follows:

$$a_m = \begin{cases} 1, & \text{if } R_m = VL \\ 2, & \text{if } R_m = L \\ 3, & \text{if } R_m = M \\ 4, & \text{if } R_m = H \\ 5, & \text{if } R_m = VH. \end{cases} \quad (2)$$

Similarly, the upper and lower bounds of the triangular number are set based on the linguistic value of the uncertainty of each answer as reported by the expert, i.e., the value of the linguistic variable  $R_c$ , as follows:

$$a_u = a_m + \sigma \quad (3)$$

$$a_l = a_m - \sigma \quad (4)$$

where

$$\sigma = \begin{cases} 0.1, & \text{if } R_c = C \\ 0.5, & \text{if } R_c = D \\ 1, & \text{if } R_c = U. \end{cases} \quad (5)$$

The range  $Q$  of the fuzzy triangular number is given by the following formula:

$$Q = a_u - a_l = 2 \times \sigma. \quad (6)$$



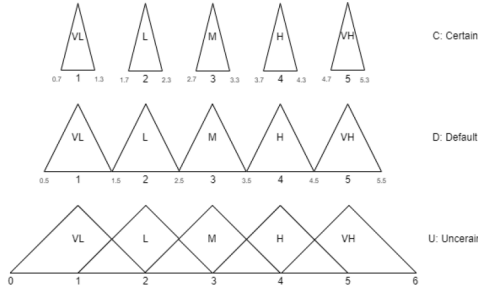


Fig. 3. Schematic representation of the fuzzy numbers mapped to the linguistic expert judgments via the fuzzifier.

From the above equations, it is obvious that the main answer of the expert (i.e.,  $R_m$ ) defines the center of the fuzzy triangular number, whereas the uncertainty (i.e.,  $R_c$ ) defines its range. The more certain the response of the expert, the narrower the triangle. The more uncertain the response of the expert, the wider the triangle. The values of  $\sigma$  given in (5) were defined based on the authors' expertise. Of course, they can be defined differently if considered necessary.

In Fig. 3, the various fuzzy triangular numbers that can be produced through the fuzzification unit are illustrated. In fact, based on the linguistic values that are provided by the experts, a fuzzy triangular number from the list presented in Fig. 3 is assigned to each one of the responses.

The reasoning behind the selection of the triangular fuzzy sets, instead of other membership functions is twofold: 1) they form the simplest transition from scalar numbers, and, thus, they are easy to comprehend even by experts with no relevant mathematical background; and 2) they are highly suitable for our model, as the center corresponds to the main answer (akin to a 5-point Likert scale) and the width captures judgment uncertainty.

### C. The HEM

The main element of the proposed approach (as can be seen in Fig. 1) is the HEM, which is responsible for computing the LoT of a given AI system, based on the expert judgments (i.e., assessments). The internal structure of the proposed HEM is illustrated in Fig. 4. As can be seen, the overall Trust of an AI system is hierarchically decomposed into  $N$  PoT (see Section II). These pillars are further decomposed into low-level criteria (i.e., questions) that are closely associated with the defined pillars. The proposed model takes as input the fuzzy expert judgments and aggregates them to compute the overall LoT of the analyzed AI system. The computation of LoT is formally described as follows.

Suppose that  $F$  corresponds to the LoT of a given AI system or application that is computed by a HEM with  $N$  PoT. Let us assume that  $g_i$  is the weight of the  $i$ th pillar, denoting the relative importance of this pillar for the overall LoT of the analyzed AI system, and  $s_i$  corresponds to the score of the  $i$ th pillar as computed from the responses of the experts to the low-level criteria of the model. Both  $g_i$  and  $s_i$  are fuzzy triangular numbers of the form  $g_i = (g_i^l, g_i^m, g_i^u)$  and  $s_i = (s_i^l, s_i^m, s_i^u)$ .

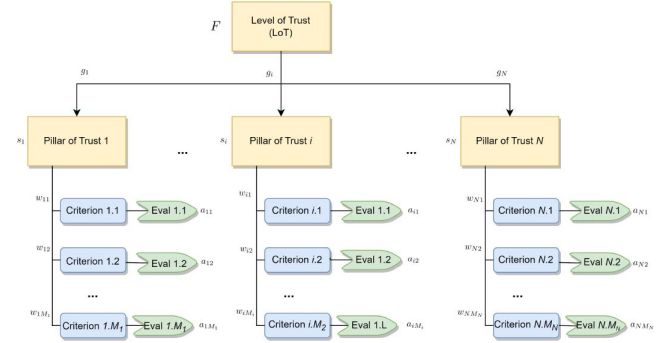


Fig. 4. General structure of the HEM for evaluating the trustworthiness of AI systems.

Based on these definitions, the LoT  $F$  is a fuzzy triangular number, given by the following formula:

$$F = \sum_{i=1}^N g_i s_i = \left( \sum_{i=1}^N g_i^l s_i^l, \sum_{i=1}^N g_i^m s_i^m, \sum_{i=1}^N g_i^u s_i^u \right). \quad (7)$$

As already mentioned, the  $s_i$  of the  $i$ th pillar is computed from the fuzzy triangular responses of the experts to the low-level criteria that are associated with the  $i$ th pillar. Let  $a_{ij}$  be the fuzzy answer to the  $j$ th question of the  $i$ th high-level criterion. It is a fuzzy triangular number of the form  $a_{ij} = (a_{ij}^l, a_{ij}^m, a_{ij}^u)$  (see Section III-B). The score  $s_i$  of the  $i$ th pillar is a fuzzy number given by the following formula:

$$s_i = \sum_{j=1}^{M_i} w_{ij} a_{ij} = \left( \sum_{j=1}^{M_i} w_{ij}^l a_{ij}^l, \sum_{j=1}^{M_i} w_{ij}^m a_{ij}^m, \sum_{j=1}^{M_i} w_{ij}^u a_{ij}^u \right) \quad (8)$$

where  $M_i$  is the total number of the low-level criteria that are associated with the  $i$ th pillar, and  $w_{ij}$  is the impact of the  $j$ th low-level criterion on the  $i$ th pillar of the model. It is a fuzzy triangular number of the form  $w_{ij} = (w_{ij}^l, w_{ij}^m, w_{ij}^u)$ .

Hence, based on (8), the complete formula for computing the LoT  $F$  of a given AI system based on the expert judgments is provided as follows:

$$\begin{aligned} F &= \sum_{i=1}^N g_i \left( \sum_{j=1}^{M_i} w_{ij} a_{ij} \right) = \\ &= \left( \sum_{i=1}^N g_i^l \left( \sum_{j=1}^{M_i} w_{ij}^l a_{ij}^l \right), \sum_{i=1}^N g_i^m \left( \sum_{j=1}^{M_i} w_{ij}^m a_{ij}^m \right), \right. \\ &\quad \left. \sum_{i=1}^N g_i^u \left( \sum_{j=1}^{M_i} w_{ij}^u a_{ij}^u \right) \right). \end{aligned} \quad (9)$$

From the aforementioned equations it is clear that the LoT  $F$  of a given AI system is computed by taking the fuzzy weighted average of the scores of the  $N$  selected PoT, which, in turn, are computed by taking the fuzzy weighted average of the expert judgments (i.e., responses) to the low-level criteria (i.e., questions) that are associated to each pillar. Hence, the expert

TABLE II  
MAPPING BETWEEN THE CRISP VERSION OF THE  
LEVEL OF TRUST (LoT) THAT IS GENERATED BY  
THE HIERARCHICAL EVALUATION MODEL (HEM)  
AND ITS LINGUISTIC RANKS

| LoT (Crisp Value) | Stars | Linguistic Rank |
|-------------------|-------|-----------------|
| [4.5, 5]          | ***** | Very High       |
| [3.5, 4.5)        | ****  | High            |
| [2.5, 3.5)        | ***   | Medium          |
| [1.5, 2.5)        | **    | Low             |
| < 1.5             | *     | Very Low        |

judgments are aggregated through several layers of aggregation to compute the LoT of the system under analysis.

#### D. Defuzzification

Since the final LoT  $F$  that is produced by HEM is a fuzzy triangular number, a defuzzification step is required in order to determine its crisp numerical (i.e., scalar value), as fuzzy numbers are difficult to understand by stakeholders without a relevant mathematical background. To this end, we use the center of mass of the fuzzy triangular number in order to defuzzify its value. The center of mass  $x^*$  of a fuzzy number  $x$  with membership function  $\mu_C(x)$  is given by the formula:

$$x^* = \frac{\int \mu_C(x)xdx}{\int \mu_C(x)dx}. \quad (10)$$

We decided to use the center of mass because the final crisp value is derived based on both the center of the fuzzy triangular number and its width. Hence, the uncertainty of the fuzzy triangular number is taken into consideration for the computation of the final crisp value. The crisp/scalar version of the LoT of a given application is a continuous number that resides in the  $[1, 5]$  interval. To facilitate understanding, apart from the crisp value of the LoT, we assign a linguistic rank denoting how trustworthy the analyzed AI system is. These ranks, along with the ranges of the crisp values, are illustrated in Table II and were defined based on the author's expertise but can be redefined by experts.

#### E. Certification

The final step of the proposed methodology is the certification, which involves a binary decision on whether the analyzed AI system could receive a certificate of trust. The decision is based on the results that are provided by HEM (Section III-C). The certification process is formally described in Algorithm 1.

As shown in Algorithm 1, a certificate is awarded to an AI system if its LoT  $F$  is above a threshold  $T$ , and the scores of all the PoT that are marked as critical (if any) are higher than pillar-specific thresholds. These thresholds are defined by the experts based on the application itself. Therefore, the overall approach is highly configurable and adaptable. This enables the experts to define specific thresholds for the overall LoT and for the individual PoT, in case some of the pillars are critical for a given application or application domain. For instance, in life-critical applications, safety is the most critical attribute, and therefore, a certificate should be issued only if the Safety criterion is found to be very high, regardless of

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#### Algorithm 1: Trustworthiness Certification Algorithm.

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**Input:** Level of Trust  $F$  of the analyzed AI system

**Input:** Set  $S$  of scores  $s_1, s_2, \dots, s_M$  of  $M$  critical high-level trustworthiness criteria

**Input:** Threshold  $T$  for overall trustworthiness

**Input:** Thresholds  $t_1, t_2, \dots, t_M$  for  $M$  critical high-level criteria

**Output:** Certificate of Trust (if awarded)  
**or** Report (if not awarded)

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**if**  $F > T$  **and** ( $S = \emptyset$  **or** ( $S \neq \emptyset$  **and**  $s_j > t_j$  for all  $j \in \{1, 2, \dots, |S|\}$ )) **then**

    Award the Certificate of Trust;

**Return** the Certificate of Trust;

**end**

**else**

    Generate Issues Report;

**Return** the report;

**end**

**Return**  $F$  and the Certificate of Trust (if awarded);

---

whether the overall LoT of the analyzed system is found to be satisfactory. If the trustworthiness of the analyzed AI system is high, the methodology produces a certificate of trust. Otherwise, a detailed Report with the identified issues that need to be fixed is presented to the users.

#### IV. MODEL CALIBRATION—FUZZY ANALYTIC HIERARCHY PROCESS (FAHP)

The trustworthiness evaluation model should be highly configurable and adaptable, allowing the generation of custom models tailored for specific application domains and regulatory frameworks. The main parameters of HEM are its weights, which denote the relative importance of the high- and low-level criteria of trust. The weights cannot be data-driven and should be defined by experts.

The elicitation of the model's weights is a difficult task due to the interdependencies that exist among the various criteria of the model. The proposed HEM can be viewed as a multicriteria evaluation model, rendering MCDM techniques highly suitable for eliciting its weights. Among the MCDM techniques, the analytic hierarchy process (AHP) [16] is the most widely used for weights elicitation [17], [18]. According to AHP, the experts compare the relative importance of two criteria at a time, and construct a pairwise comparison (PWC) matrix with their judgments, the eigenvalues and eigenvectors of which are utilized to compute a set of weights that reflect the expert judgments. AHP allows the experts to focus on a couple of criteria, and the interdependencies (i.e., cross impacts) are automatically considered by the mathematical equations and reflected in the produced weights.

However, since AHP does not take into account the uncertainty of expert judgments and can produce only scalar weights, it is not suitable for calibrating our model. To this end, we provide a Fuzzy AHP (FAHP) approach, which is based on

TABLE III  
GENERAL STRUCTURE OF THE PAIRWISE COMPARISON (PWC) MATRIX  
USED FOR CALCULATING THE WEIGHTS OF  $N$  PILLARS, NAMELY  
 $P_1, P_2, \dots, P_N$

| $F$      | $P_1$    | $P_2$    | ... | $P_j$    | ... | $P_N$    |
|----------|----------|----------|-----|----------|-----|----------|
| $P_1$    | $e_{11}$ | $e_{12}$ | ... | $e_{1j}$ | ... | $e_{1N}$ |
| $P_2$    | $e_{21}$ | $e_{22}$ | ... | $e_{2j}$ | ... | $e_{2N}$ |
| $\vdots$ | $\vdots$ | $\vdots$ |     | $\vdots$ |     | $\vdots$ |
| $P_i$    | $e_{i1}$ | $e_{i2}$ | ... | $e_{ij}$ | ... | $e_{iN}$ |
| $\vdots$ | $\vdots$ | $\vdots$ |     | $\vdots$ |     | $\vdots$ |
| $P_N$    | $e_{N1}$ | $e_{N2}$ | ... | $e_{Nj}$ | ... | $e_{NN}$ |

Note: The value of  $e_{ij}$  denotes how stronger (or weaker) the impact of  $P_i$  on the overall trust is, compared to the corresponding impact of  $P_j$ , according to the experts.

fuzzy triangular representations of the expert judgments and allows the derivation of fuzzy triangular weights. A formal definition of the FAHP is provided as follows. For reasons of brevity, we demonstrate the process for the case of deriving the weights of the PoT. The same process can be applied to derive the weights of the low-level criteria.

Suppose that we have a model with  $N$  PoT  $P_1, P_2, \dots, P_N$ . These PoT are not equally important for the associated software application (see Fig. 4). Suppose that we have the fuzzy PWC matrix  $A = (e_{ij})_{N \times N}$ , where its elements  $e_{ij}$  are fuzzy triangular numbers of the form  $e_{ij} = (e_{ij}^l, e_{ij}^m, e_{ij}^u)$ , where  $e_{ij}^l$ ,  $e_{ij}^m$ , and  $e_{ij}^u$  denote the relative importance of  $P_i$  and  $P_j$  for the application. According to Saaty's original suggestion,  $e_{ij}^l, e_{ij}^m, e_{ij}^u \in [1, (1/9)] \cup [1, 9]$  [16]. We wish to estimate a set of positive fuzzy weights  $G = g_1, g_2, \dots, g_N$ , such that

$$e_{ij} \simeq \frac{g_i}{g_j}. \quad (11)$$

The fuzzy PWC matrix is shown in Table III.

The fuzzy comparison matrix  $A$  should be a fuzzy positive reciprocal matrix, which should also be fuzzily consistent. The definitions of the fuzzy positive reciprocal matrix and the fuzzily consistent reciprocal matrices are provided as follows.

**Definition 2:** A fuzzy matrix  $A = (e_{ij})_{N \times N}$  is a positive reciprocal matrix if

$$e_{ij} \times e_{ji} \simeq 1 \text{ and } e_{ij}^m \times e_{ji}^m = 1, \forall i, j = 1, 2, \dots, N$$

where  $\simeq$  means approximately equal.

**Definition 3:** A positive reciprocal matrix  $A = (e_{ij})_{N \times N}$  is fuzzily consistent if

$$e_{ij} \times e_{jk} \simeq e_{ik} \text{ and } e_{ij}^m \times e_{jk}^m = e_{ik}^m, \forall i, j = 1, 2, \dots, N$$

where  $\simeq$  means approximately equal.

In other words, Definition 2 requires an exact consistency in the modes and an approximate consistency in the spreads (ranges). However, the approximate consistency can be interpreted differently in different circumstances. Since FAHP requires a fuzzily consistent reciprocal matrix, the elicitation process can require reviewers to revise pairwise judgments when incoherence is indicated via consistency checks, thus reducing the chance of producing inconsistent weights.

Since the fuzzy PWC matrix is a reciprocal matrix, the experts need to define only half of its values. For instance, the experts could complete the upper triangular part of  $A$  with their fuzzy judgments, and the lower triangular part would be completed automatically according to  $e_{ij} = 1/e_{ji}, \forall i, j = 1, 2, \dots, N$ , and the fuzzy inverse arithmetic operation proposed by the Zadeh principle [19].

After completing the fuzzy PWC matrix  $A$ , the related normalized set of positive fuzzy weights can be derived based on the fuzzy logarithmic least squares method (FLLSM), which was proposed by van Laarhoven and Pedrycz [15] and Boender et al. [20]. According to the proposed method, the final set of fuzzy weights is the set of weights that minimizes the following objective function:

$$\sum_{i=1}^N \sum_{j=i+1}^N \left( \ln(e_{ij}) - \ln\left(\frac{g_i}{g_j}\right) \right) \rightarrow \min \quad (12)$$

which can be transformed into the minimization of the fuzzy version of the logarithmic regression function:

$$\begin{aligned} \sum_{i=1}^N \sum_{j=i+1}^N \{ & \ln(e_{ij}^l) - \ln(g_i^l) + \ln(g_j^u) \}^2 \\ & + \{ \ln(e_{ij}^m) - \ln(g_i^m) + \ln(g_j^m) \}^2 \\ & + \{ \ln(e_{ij}^u) - \ln(g_i^u) + \ln(g_j^l) \}^2. \end{aligned} \quad (13)$$

According to Boender et al. [20] and Buckley [21], the solution of the optimization problem is expected to converge to the geometric mean. Hence, the weights  $g$  obtained by this method have the following lower, modal, and upper values:

$$(g_i^l, g_i^m, g_i^u) = \left( \frac{\prod_{j=1}^N (e_{ij}^l)^{1/N}}{\sum_{i=1}^N \prod_{j=1}^N (e_{ij}^u)^{1/N}}, \frac{\prod_{j=1}^N (e_{ij}^m)^{1/N}}{\sum_{i=1}^N \prod_{j=1}^N (e_{ij}^m)^{1/N}}, \frac{\prod_{j=1}^N (e_{ij}^u)^{1/N}}{\sum_{i=1}^N \prod_{j=1}^N (e_{ij}^l)^{1/N}} \right). \quad (14)$$

The described weights elicitation process guarantees that the produced fuzzy weights given by (14) are normalized, that is, their fuzzy geometric mean is approximately equal to one. The same process is followed for computing the weights  $w_{ij}$  of the PoT.

## V. DEMONSTRATION

In this section, the proposed methodology is demonstrated in a real-world setting through its application for evaluating the trustworthiness of two real-world AI systems. These AI systems are proprietary systems that have been developed by CErTH and are used for predicting the energy consumption of the ITI Smart Home, which is located at CErTH's premises and serves as a testbed in various EU research projects.

For this purpose, a dedicated evaluation model, i.e., the base model, was constructed based on the EU's regulatory framework (known as the AI Act [2]) and used for the evaluation of the AI systems. In brief, the base model relies on the main PoT that were defined in Section II. The low-level criteria (i.e.,

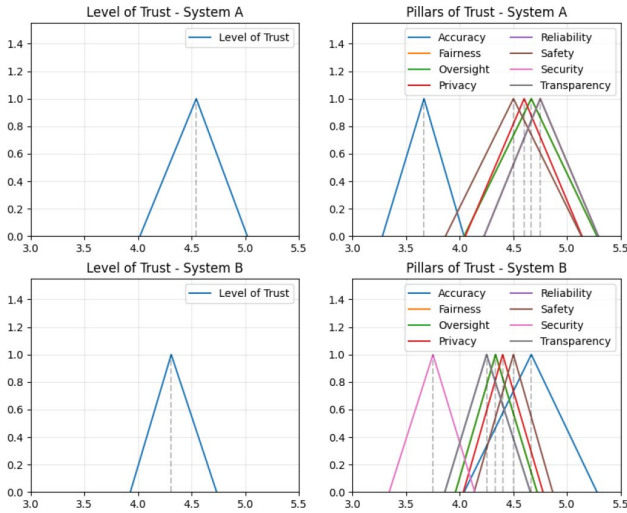


Fig. 5. Detailed results of the AI trust evaluation tool for System A (top) and System B (bottom).

the questions that should be answered by the experts) for each PoT are presented in Table IV, which can be found in the supplementary material (Appendix) of the article. These low-level criteria were defined based on the AI Act [2] and its associated HLEG report [6].

To facilitate understanding and enable the utilization of the base model in practice, a prototype tool has been implemented and is available online.<sup>4</sup> In the future, we are planning to implement the whole methodology in the form of a toolkit, which will enable the construction of custom evaluation models (similar to the base model) that can be used in practice for evaluating and certifying the trustworthiness of AI systems.

The two AI systems were analyzed by the authors based on their expertise and the guidelines provided by the AI Act. Particularly, the authors provided their responses to the questions of the questionnaire. Then the model was executed to compute the scores of the PoT and the overall LoT of the analyzed systems. Fig. 5 presents the evaluation results.

As can be seen in Fig. 5, the crisp value (i.e., defined by the center of mass) of the LoT of the two AI systems is found to be 4.5 and 4.3, respectively, which correspond to very high and high LoT, respectively. Since no critical PoT was defined, and the value of the LoT was higher than  $T$ , a threshold defined by the experts, particularly for the domain of the use case, both can receive a certificate of trust.

The evaluation results can also be utilized for comparing the two AI systems with respect to their PoT. A comparison of their crisp scores, computed based on (10), is illustrated in Fig. 6. As can be seen from this figure, the first AI system ranks much better with respect to the Security pillar, whereas the second one has a higher score in the Accuracy pillar. Therefore, the experts can choose the AI system that better fits their needs. If the security is considered more important, the first one could be selected; otherwise, the second one would provide more accurate predictions. Therefore, the proposed methodology could also be used for comparing different AI systems with respect to

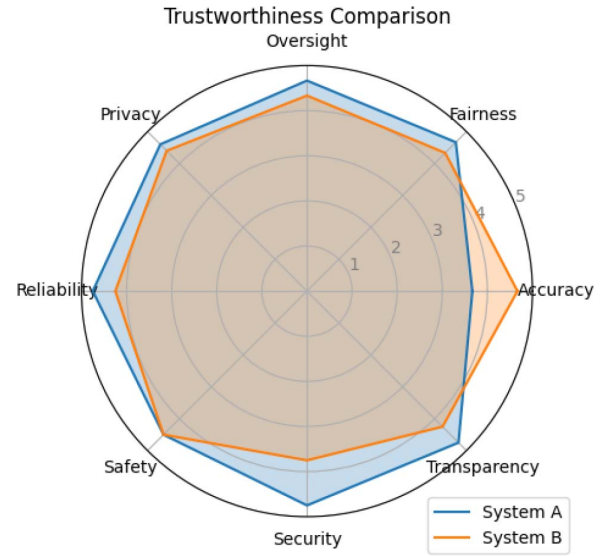


Fig. 6. Radar chart comparing the PoTs of the AI systems.

their PoT, and selecting the one that better satisfies the criteria that are considered more important in a given use case. The detailed low-level assessments of the experts are provided in the supplementary material (Appendix C) of the article.

It should be noted that the derivation of the “base model” questionnaire shown in Table IV followed a structured approach based on the AI Act [2], the HLEG guidelines on trustworthy AI [6], and the ALTAI framework [4]. The authors individually analyzed these materials and, using the goal question metric (GQM) method [22], identified candidate questions for each one of the selected PoT. Then, a focus group was held among the authors, in which the candidate questions were discussed and refined, and overlapping and similar questions were merged, leading to the final list of candidate questions. The authors voted for the five most relevant questions per PoT, and those questions that achieved a unanimous consensus (i.e., were selected by all the authors) were included in the final questionnaire. Hence, the involvement of three independent authors, the GQM approach, the focus group, and the consensus voting acted as validation mechanisms for ensuring the quality of the derived list. The limitation of a maximum of five questions per pillar was set to ensure the manageability of the assessment procedure and the interpretability of its results. A more extensive questionnaire was not considered necessary, due to the fact that it was meant only for demonstration purposes of the TrustAI approach.

## VI. BIAS FROM MULTIPLE EXPERTS

Although the proposed TrustAI methodology is meant for assessments performed by a single expert, it is reasonable to expect organizations to decide to involve multiple experts, either during the evaluation process or during the model calibration phase. Fusing judgments by multiple experts comes with the risk of aggregating bias and subjectivity, an important issue that could potentially undermine the reliability of the assessment results. Hence, the involvement of multiple experts must be treated with care, and various mitigation techniques need to be considered.

<sup>4</sup><http://160.40.52.130:3010/>



First, concrete and transparent selection criteria should be defined, and only experts who fully satisfy these criteria should be involved in the process. More specifically, experts should be selected at least based on their proven domain expertise, years of experience, prior record of relevant review involvement, and lack of relevant conflicts of interest. The organization can freely define its selection criteria of choice, but it is important that these criteria are formally recorded and accessible openly or upon request.

Second, to mitigate and reduce potential bias, the experts should restrict themselves only to the potential list of answers that are supported by the TrustAI methodology, i.e., main answer ( $R_m$ ) and confidence ( $R_c$ ) as described in Section III-A, and follow the assessment guidelines that are provided by the platform to provide their answers (see Appendix B in the supplementary material of the article). By doing so, the possibility of resulting in significantly different judgments (i.e., assessments) is reduced.

Third, inter-rater reliability checks should be performed, both at the level of the expert judgments for the individual low-level criteria (i.e., questions) and at the level of the broader pillars. The inter-rater reliability can be computed on the ordinal mapping of linguistic answers (1–5) or on defuzzified values of the fuzzy scores using standard agreement statistics (e.g., Krippendorff's  $\alpha$ ). In case of low agreement (i.e., strong disagreement), Delphi-style consensus rounds can be applied to resolve inconsistencies, whereas the final assessment could be accompanied by dispersion indicators (e.g., variance across experts) for reasons of transparency.

All the aforementioned steps aim to alleviate bias arising from the involvement of multiple experts. However, subjectivity and uncertainty in expert judgments can only be reduced, not eliminated (a known undecidable problem [23]). Thus, TrustAI incorporates techniques for effectively reducing such bias (e.g., structured judgments, modeling confidence with fuzzy logic, and post-hoc multiexpert reliability evaluation) to provide more reliable and credible assessments.

## VII. CONCLUSION

In this article, we proposed an expert-based methodology to evaluate and certify the trustworthiness of AI systems. The methodology is based on the hierarchical decomposition of the notion of trust into a set of pillars of trust and low-level criteria, which are assessed by experts. Fuzzy logic is utilized to model the uncertainty of expert judgments, whereas a fuzzy MCDM approach is provided for enabling the generation of custom assessment models. The methodology was demonstrated through a use case on two real-world AI systems, based on a base model and a prototype tool. Future work includes the implementation of a dedicated toolkit enabling the construction of custom evaluation and certification models, and the conduction of a broader expert survey to construct domain- and application-specific models.

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